# Momentum in two dimensions

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Conserve momentum separately along perpendicular axes.

**SUCCESS CRITERIA - I can...**

1. Choose and label two axes.
2. Add momentum components along matching directions.
3. Connect component sums to the final motion.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

**LANGUAGE GOAL** Use “component” and “system” to explain your model or result. Sentence starters are optional.

**component**: the part of a vector along one chosen axis

**system**: the object or objects chosen for analysis

**conserved**: unchanged in total under the stated conditions

## 01 Bring back an earlier idea

Initial momentum is 0.3 kg·m/s and measured final momentum is 0.29 kg·m/s.
**Find** percent deviation.
**Explain** why this is not itself a measured uncertainty.

**Word help:** momentum = mass times velocity, including direction

**Sentence starter:** Under the assumption ..., the evidence ... supports ... because ... .

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $p_{x,i} = p_{x,f}$; $p_{y,i} = p_{y,f}$; $p_x = mv_x$; $p_y = mv_y$.
Use total system momentum in each direction; negligible external impulse. CPA components are supplied.

**Worked reasoning:** Two objects have momenta (3, 0) and (0, 4) kg·m/s. Their total is (3, 4), magnitude 5 kg·m/s. Add components before finding the magnitude.

**Your turn: Draw** a tip-to-tail momentum diagram and label each axis with units.

**Sentence starter:** The ... in my model represents ... . Its label shows ... .

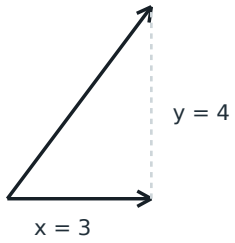
**Marzano check - circle:** ① ② ③

## 03 Represent and interpret

Two incoming arrows point east and north.
**Sketch** the direction of their vector sum. Can their ordinary lengths be added for the diagonal magnitude?

**Word help:** magnitude = size of a quantity without its direction

**Sentence starter:** The ... in my model represents ... . Its label shows ... .




The arrows represent horizontal and vertical components; use their directions to describe the sum.

**Marzano check - circle:** ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

## 04 Solve and justify

A 2 kg cart at 3 m/s east sticks to a 1 kg cart at 4 m/s north.
**Find** final velocity components, speed and angle north of east.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3

## 05 Analyze the reasoning

Why must both momentum component sums use the same system and same time interval?

**Sentence starter:** Under the assumption …, the evidence … supports … because … .

**Marzano check - circle:** 1 2 3

## 06 Choose a method

Two ideal Sun-orbits have $a_B = 4\ a_A$.
**Find** $T_B/T_A$.
**Explain** why the same normalized Sun-orbit equation is not automatically valid around another body.

**Word help:** orbit = a path around another body under gravity

**Sentence starter:** Under the assumption …, the evidence … supports … because … .

**Marzano check - circle:** 1 2 3

## 07 Exit check

Initial total momentum is(10, 0) kg·m/s. One fragment leaves with (4, 3).
**Find** the other momentum vector and its magnitude.

**Word help:** magnitude = size of a quantity without its direction

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: Apply it another way

A 2 kg cart moves east at 3 m/s and a 1 kg cart north at 4 m/s before sticking.
**Find** final velocity components, speed and direction.

**Sentence starter:** I found … [unit]. This value means … in this situation.

**Marzano check - circle:** 1 2 3